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Systems and methods for thrombus aspiration

Abstract

Systems and methods include positioning of a catheter lumen in a first position with respect to a thrombus, the catheter lumen defined by a catheter, initiating of evacuation of a tubing lumen, where the tubing lumen is not in fluid communication with the catheter lumen, determination that a pressure within the tubing lumen is equal to or less than a target aspiration pressure, and, in response to the determination that the pressure within the tubing lumen is equal to or less than a target aspiration pressure, automatic establishment of fluid communication between the tubing lumen and the catheter lumen.

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Background/Summary

BACKGROUND

(1) As used herein, the term elongated medical device (EMD) refers to, but is not limited to, catheters (e.g., guide catheters, microcatheters, aspiration catheters, balloon/stent catheters), wire-based devices (e.g., guidewires, microwires, embolization coils, stent retrievers, etc.), and medical devices comprising any combination of the above. Generally, EMDs may be used for many minimally-invasive medical procedures. Such procedures may facilitate the diagnosis and treatment of diseases of various vascular systems, and include neurovascular interventional (NVI) (or neurointerventional) surgery, percutaneous coronary intervention (PCI), and peripheral vascular intervention (PVI). Generally, these procedures involve navigating a guidewire through patient vasculature, and advancing a working catheter via the guidewire to deliver therapy.

(2) A catheterization procedure starts by gaining access into an appropriate vessel, such as an artery or vein, by inserting a sheath therein. Next, for example, a diagnostic guidewire is advanced within the sheath to a primary location such as an internal carotid artery (in NVI), a coronary ostium (in PCI) or a superficial femoral artery (in PVI). A guide catheter is then advanced over the diagnostic guidewire to the primary location. The diagnostic guidewire is removed, and a guidewire suitable for navigating the target vasculature is then pushed through the guide catheter to a target location (e.g., lesion, thrombus) within the vasculature. In certain situations, such as in the case of tortuous anatomy, a support catheter or microcatheter is inserted over the guidewire to assist in navigating the guidewire therethrough.

(3) In order to navigate the guidewire or guidewire/microcatheter toward the target location, a physician advances the guidewire or guidewire/microcatheter while manipulating its proximal end so as to direct the distal tip into appropriate vessel branches along the path to the target location, while avoiding advancing into other vessel branches. Prior to the procedure, the physician may use an imaging system (e.g., a fluoroscope) to obtain successive contrast-enhanced images of the patient vasculature and may select one of the images for use as a roadmap to navigate the guidewire or guidewire/microcatheter to the target location. Contrast-enhanced images are also obtained while the physician navigates the guidewire or guidewire/microcatheter so that the physician can verify that the device is moving along the correct path to the target location.

(4) Robotic catheter-based procedure systems facilitate the above processes by supporting each of the guidewires and catheters and physically manipulating each of the guidewires and catheters as desired by a physician. For example, a robotic catheter-based procedure system may hold a guidewire and include mechanical components to advance, retract and rotate the guidewire in response to physician commands. The physician may provide such commands via input devices (e.g., joysticks, buttons, scroll wheels, touch screens) mounted to a control console. The input devices may allow the physician to select one or more guidewires and/or catheters to be controlled at a given time. Robotic catheter-based procedure systems may thereby provide greater control and accuracy than purely manual catheterization procedures. Moreover, the control console and thus the physician may be shielded from x-rays emitted by the imaging system used to track the locations of the guidewires and/or catheters during the procedures.

(5) Robotic catheter-based procedure systems may be used to assist a physician in performing catheterization procedures such as, for example, NVI, PCI and PVI. Examples of robotic-assisted NVI procedures include coil embolization of aneurysms, liquid embolization of arteriovenous malformations and mechanical thrombectomy of large vessel occlusions in the case of acute ischemic stroke. In such NVI procedures, a physician uses a robotic system to gain lesion access by manipulating a neurovascular guidewire and microcatheter to deliver therapy which restores normal blood flow. Access is provided using a sheath or guide catheter as described above but may also require an intermediate catheter to provide additional distal territory and/or to provide adequate support for the microcatheter and guidewire. The distal tip of a guidewire is navigated into or past the lesion depending on the type of lesion and treatment. For treating aneurysms, the

microcatheter is advanced into the lesion, the guidewire is removed, and several coils are deployed into the aneurysm through the microcatheter. The coils are then used to embolize the aneurysm. For treating arteriovenous malformations, a liquid embolic is injected into the malformation via the microcatheter.

(6) In conventional mechanical thrombectomy, aspiration may be used to treat vessel occlusions. Aspiration may be performed directly through the aforementioned microcatheter or using a larger-bore aspiration catheter. Once the microcatheter or aspiration catheter has been navigated to the occluding thrombus, an aspiration pump connected thereto is activated to generate negative pressure which causes the thrombus to be pulled into the catheter. Conventional aspiration systems may require significant time to completely remove a thrombus from a vessel. Additionally, conventional systems may generate and apply the negative pressure longer than is required to remove the thrombus. Either of these shortcomings increases risk of injury to the vasculature and/or to the aspiration pump. Systems are desired to improve the efficiency, efficacy and/or safety of mechanical thrombectomy using robotic catheter-based procedure systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a robotic catheter-based procedure system in accordance with some embodiments.
- (2) FIG. 2 is a schematic block diagram of a robotic catheter-based procedure system in accordance with some embodiments.
- (3) FIG. 3 is a perspective view of a robotic drive for a robotic catheter-based procedure system in accordance with an embodiment.
- (4) FIG. 4 is a schematic diagram of a thrombectomy system according to some embodiments.
- (5) FIG. 5 is a schematic diagram of a thrombectomy system according to some embodiments.
- (6) FIG. 6 is a schematic diagram of a thrombectomy system according to some embodiments.
- (7) FIG. 7 is a schematic diagram of a thrombectomy system according to some embodiments.
- (8) FIG. 8 is a schematic diagram of a thrombectomy system according to some embodiments.
- (9) FIG. 9 is a flow diagram of a thrombectomy process according to some embodiments.
- (10) FIG. 10 is a cutaway diagram of a vessel exposing stacked EMDs during a thrombectomy process according to some embodiments.
- (11) FIG. 11 is a flow diagram of a thrombectomy process according to some embodiments.
- (12) FIGS. 12a and 12b comprise a flow diagram of a thrombectomy process according to some embodiments.
- (13) FIGS. 13a and 13b comprise a flow diagram of a thrombectomy process according to some embodiments.

DETAILED DESCRIPTION

- (14) The following description is provided to enable any person in the art to make and use the described embodiments. Various modifications will remain apparent to those in the art.
- (15) According to some embodiments, aspiration of a thrombus is initiated only after a determination that an aspiration (i.e., vacuum) source has reached a threshold negative pressure. Such initiation increases removal effectiveness, resulting in faster removal of the thrombus in many instances.
- (16) In one example, an aspiration catheter lumen defined by an aspiration catheter is positioned in a first position with respect to a thrombus. Evacuation is then initiated of a tubing lumen which is not in fluid communication with the aspiration catheter lumen. After a period of time, it is determined that a pressure within the tubing lumen is equal to or less than a target aspiration pressure. In response to the determination that the pressure within the tubing lumen is equal to or

less than a target aspiration pressure, fluid communication is automatically established (e.g., by opening a valve, by releasing a clamp) between the tubing lumen and the aspiration catheter lumen. Such a process may serve to evacuate the thrombus more quickly than otherwise. In some aspects, an operator is informed that the pressure within the tubing lumen is equal to or less than the target aspiration pressure, and the operator transmits a command to automatically establish fluid communication between the tubing lumen and the aspiration catheter lumen.

(17) Embodiments may further determine whether the pressure within the tubing lumen is greater than a second pressure and, if it is determined that the pressure within the tubing lumen is greater than the second pressure, automatically terminate (e.g., by closing a valve, by engaging a clamp) fluid communication between the tubing lumen and the aspiration catheter lumen. These aspects may minimize aspiration time to that which is actually needed to remove the thrombus.

(18) Similarly, some aspects include presentation of an indication to an operator that the pressure within the tubing lumen is greater than the second pressure. After presenting the indication, and while the pressure within the tubing lumen is greater than the second pressure, fluid communication between the tubing lumen and the aspiration catheter lumen is automatically terminated in response to receipt of a command from the operator.

(19) Embodiments may also include repositioning the aspiration catheter as necessary during the procedure to maintain a suitable position of the tip of the aspiration catheter relative to the face of the thrombus. For example, after automatically establishing fluid communication between the tubing lumen and the aspiration catheter, and before determining whether the pressure within the tubing lumen is greater than a second pressure, it is determined whether the pressure within the tubing lumen is within a predefined range. If it is determined that the pressure within the tubing lumen is within the predefined range, the position of the aspiration catheter lumen is automatically adjusted until the pressure within the tubing lumen is less than a threshold aspiration pressure.

(20) FIG. 1 is a perspective view of catheter-based procedure system **10** in accordance with some embodiments. Catheter-based procedure system **10** may be used to perform catheter-based medical procedures, e.g., percutaneous intervention procedures such as PCI (e.g., to treat STEMI), NVI (e.g., to treat an emergent large vessel occlusion (ELVO)), and PVI (e.g., for critical limb ischemia (CLI)). Catheter-based medical procedures may include diagnostic catheterization procedures during which one or more catheters or other EMDs are used to aid in the diagnosis of a patient's disease. In one example, a contrast media may be injected into one or more arteries through a catheter and an image of the patient's vasculature is acquired while the contrast media resides therein.

(21) Catheter-based medical procedures may also include catheter-based therapeutic procedures (e.g., angioplasty, stent placement, treatment of peripheral vascular disease, thrombus removal, arterial venous malformation therapy, treatment of aneurysm) in which a catheter (or other EMD) is used to treat a disease. The particular type or nature of EMD used in a catheter-based medical procedure is selected based on the type of procedure that is to be performed. Catheter-based procedure system **10** can perform any number of catheter-based medical procedures using adjustments necessary to accommodate the specific EMDs to be used in the procedures.

(22) Catheter-based procedure system **10** includes, among other elements, bedside unit **20** and control station **26**. Bedside unit **20** includes robotic drive **24** and positioning system **22** that are located adjacent to patient **12**. Patient **12** is supported on patient table **18**. Positioning system **22** is used to position and support robotic drive **24**. Positioning system **22** may be, for example, a robotic arm, an articulated arm, a holder, etc. Positioning system **22** may be attached at one end to, for example, a rail on the patient table **18**, a base, or a cart. The other end of positioning system **22** is attached to robotic drive **24**. Positioning system **22** may be moved out of the way (along with robotic drive **24**) to allow for patient **12** to be placed on patient table **18**.

(23) Once patient **12** is positioned on patient table **18**, positioning system **22** may be used to situate or position robotic drive **24** relative to the patient **12** for the procedure. In some embodiments,

patient table **18** is operably supported by pedestal **17**, which is secured to the floor and/or earth. Patient table **18** is able to move with multiple degrees of freedom, for example, roll, pitch, and yaw, relative to pedestal **17**. Bedside unit **20** may also include operator controls and displays (not shown). For example, such controls and displays may be located on a housing of robotic drive **24**. (24) The term front will refer herein to the side of robotic drive **24** that faces the patient **12** and away from positioning system **22**, while the term rear refers to the side of robotic drive **24** that is closest to positioning system **22**. The terms top, up, and upper refer to the general direction away from the direction of gravity and the terms bottom, down, and lower refer to the general direction in the direction of gravity.

(25) Generally, robotic drive **24** may be equipped with appropriate EMDs and associated accessories (e.g., embolization coils, liquid embolics, aspiration pumps, contrast injection systems, medicine, hemostasis valve adapters, syringes, stopcocks, inflation device, etc.) to allow operator **11** to perform a catheter-based medical procedure by operating various controls of a control system such as the controls and inputs located at the control station **26**. Bedside unit **20**, and in particular robotic drive **24**, may include any number and/or combination of components to provide bedside unit **20** with the functionality described herein.

(26) Robotic drive **24** includes a plurality of device modules **32**, each of which may be controlled to drive a respective EMD. Moreover, each of device modules **32** may be controlled to move linearly toward and away from patient **12**. In some embodiments, robotic drive **24** may control one or more of device modules **32** to feed a guidewire into a diagnostic catheter and into a guide catheter in an artery of patient **12**. The EMDs enter the body (e.g., a vessel) of patient **12** at an insertion point **16** via, for example, an introducer sheath.

(27) Bedside unit **20** is in communication with control station **26**, allowing signals generated by the controls of control station **26** to be transmitted wirelessly or via hardwire to bedside unit **20** to control various functions of bedside unit **20**, including functions of the robotic drive **24**. As discussed below, control station **26** may include a control computing system **34** (shown in FIG. 2) or be coupled to the bedside unit **20** through a control computing system **34**. Bedside unit **20** may also provide feedback signals (e.g., loads, speeds, operating conditions, warning signals, error codes, etc.) to control station **26**, control computing system **34**, or both. Communication between control computing system **34** and various components of catheter-based procedure system **10** may be provided via a communication link that may be a wireless connection, cable connections, or any other means capable of allowing communication to occur between components. Control station **26** or other similar control system may be located either local to or remote from robotic drive **24**.

(28) The term local is used to refer to the location of patient **12** and bedside unit **20**. The term remote is used to refer to locations that do not have substantially-immediate physical access to bedside unit **20** and/or patient **12**. Catheter procedure system **10** may be operated by a control station **26** at the local site, a control station **26** at a remote site, or both a local control station **26** and a remote control station **26** at the same time. At a local site, operator **11** and control station **26** are located in the same room as patient **12** and bedside unit **20** or in an adjacent room.

(29) Control station **26** (and a control computing system) at a remote site may be in communication with the bedside unit **20** and/or a control computing system at a local site using communication systems and services, for example, through the Internet. In some embodiments, the remote site and the local site are in different rooms of the same building, different buildings in the same city, different cities, or other different locations where the remote site does not provide substantially-immediate physical access to bedside unit **20** and/or patient **12**.

(30) Control station **26** generally includes one or more input systems **28** including controls configured to receive user manipulations for controlling robotic drive **24** and/or various other components or systems of catheter-based procedure system **10**. In the embodiment shown, control station **26** allows operator **11** to control bedside unit **20** to perform a catheter-based medical procedure. For example, input systems **28** may be configured to cause bedside unit **20** to perform

various diagnostic or interventional procedures using EMDs controlled by drive mechanisms of robotic drive **24** (e.g., to advance, retract, or rotate a guidewire, advance, retract or rotate a catheter, inflate or deflate a balloon located on a catheter, position and/or deploy a stent, position and/or deploy a stent retriever, position and/or deploy a coil, inject contrast media into a catheter, inject liquid embolics into a catheter, inject medicine or saline into a catheter, aspirate on a catheter).

(31) One or more input systems **28** may include one or more touch screens, joysticks, scroll wheels, and/or buttons. In addition to input systems **28**, control station **26** may use additional user controls **44** such as foot switches and microphones for voice commands, etc. Input systems **28** may be configured to instruct advancement, retraction, and/or rotation of EMDs, and activation or deactivation of various components including pumps, valves, switches, clamps, etc.

(32) One or more input systems **28** may include device selection buttons to allow operator **11** to select which of the EMDs loaded into robotic drive **24** are controlled via user manipulation of input controls of one or more input systems **28**. Automated routine buttons may be selected to enable algorithmic movements of an EMD without individual direct commands from operator **11**. In some embodiments, input systems **28** may include one or more controls or icons (not shown) displayed on a touch screen (e.g., display **30**) which, when activated, cause operation of a component of catheter-based procedure system **10**.

(33) Display **30** may be configured to display information or patient-specific data to operator **11** located at control station **26**. In some embodiments, control station **26** may include two or more displays **30**. For example, display **30** may be configured to display image data (e.g., X-ray images, MRI images, CT images, ultrasound images), hemodynamic data (e.g., blood pressure, heart rate), patient record information (e.g., medical history, age, weight), lesion or treatment assessment data (e.g., intravascular ultrasound (IVUS), optical coherence tomography (OCT), fractional flow reserve (FFR)). In addition, display **30** may be configured to display procedure-specific information (e.g., procedural checklist, recommendations, duration of procedure, catheter or guidewire position, aspiration vacuum level, volume of medicine or contrast agent delivered). Further, display **30** may be configured to display information to provide the functionalities associated with a control computing system as will be described below. Display **30** may comprise a touch screen and therefore an input device of system **10**.

(34) Catheter-based procedure system **10** also includes imaging system **14**. Imaging system **14** may be any medical imaging system usable in conjunction with a catheter-based medical procedure (e.g., non-digital X-ray, digital X-ray, CT, MRI, ultrasound). In some embodiments, imaging system **14** is a digital X-ray imaging device that is in communication with control station **26**. Imaging system **14** may include a C-arm that allows imaging system **14** to rotate partially or completely around patient **12** in order to obtain images at different angular positions relative to patient **12** (e.g., sagittal views, caudal views, anterior-posterior views). In some embodiments, imaging system **14** is a fluoroscopy system including a C-arm having an X-ray source **13** and a detector **15**, also known as an image intensifier.

(35) Imaging system **14** may be configured to acquire X-ray images of appropriate areas of patient **12** during a procedure. For example, imaging system **14** may be configured to acquire one or more X-ray images of the head to diagnose a neurovascular condition. Imaging system **14** may also be configured to acquire one or more X-ray images (e.g., real time images) during a catheter-based procedure to assist operator **11** to properly position a guidewire, guide catheter, microcatheter, stent retriever, coil, stent, balloon, etc. during the procedure. The acquired image or images may be displayed on display **30**. For example, images may be displayed on display **30** to allow operator **11** to accurately move a tip of an aspiration catheter to a position adjacent to a thrombus.

(36) A rectangular coordinate system is hereby introduced including X, Y, and Z axes. The positive X axis is oriented in a longitudinal (axial) distal direction, that is, in the direction from the proximal end to the distal end. The Y and Z axes are in a transverse plane to the X axis, with the positive Z axis oriented up, that is, in the direction opposite of gravity, and the Y axis oriented accordingly

based on right-hand rule.

(37) FIG. 2 is a block diagram of catheter-based procedure system **10** in accordance with an exemplary embodiment. Catheter-based procedure system **10** may include a control computing system **34**. Control computing system **34** may physically be, for example, part of control station **26**. Control computing system **34** may generally comprise a computer processing unit suitable to control catheter-based procedure system **10** as described herein. For example, control computing system **34** may be an embedded system, a dedicated circuit, a general-purpose processor executing program code, etc. Control computing system **34** is in communication with bedside unit **20**, communications systems and services **36** (e.g., Internet, firewalls, cloud services, session managers, a hospital network), local control station **38**, additional communications systems **40** (e.g., a telepresence system), remote control station and computing system **42**, and patient sensors **56** (e.g., electrocardiogram (ECG) devices, electroencephalogram (EEG) devices, blood pressure monitors, temperature monitors, heart rate monitors, respiratory monitors). Control computing system **34** may also be in communication with imaging system **14**, patient table **18**, additional medical systems **50**, contrast injection systems **52** and adjunct devices **54** (e.g., IVUS, OCT, FFR). (38) As described above, bedside unit **20** includes robotic drive **24**, positioning system **22**, and additional controls and displays **46**. Additional controls and displays **46** may be located on a housing of robotic drive **24**. Interventional devices and accessories **48** (e.g., guidewires, catheters) interface to bedside system **20**. In some embodiments, interventional devices and accessories **48** may include specialized devices (e.g., IVUS catheter, OCT catheter, FFR wire, diagnostic catheter for contrast) which interface to their respective adjunct devices **54** (e.g., an IVUS system, an OCT system, an FFR system).

(39) In various embodiments, control computing system **34** is configured to receive and generate control signals based on user manipulation of the controls of one or more input systems **28** of local control station **38**. Remote control station and computing system **42** may include similar components to the local control station **38**. Remote **42** and local **38** control stations can be different and tailored based on their required functionalities. Additional user controls **44** may include, for example, one or more foot input controls. A foot input control may be configured to allow an operator to select functions of imaging system **14** such as turning an X-ray source on and off and scrolling through different stored images. In another embodiment, a foot input may be configured to allow an operator to select which EMDs are mapped to which controls of input system **28**. Additional communication systems **40** (e.g., audio conference, video conference, telepresence) may be employed to assist operator interaction with the patient, medical staff (e.g., angio-suite staff), and/or equipment in the vicinity of the bedside.

(40) Control computing system **34** is also in communication with pump **58** and vacuum control **59**. Pump **58** may comprise any source for generating a vacuum suitable for thrombus aspiration. In some embodiments, control **59** may be operable to selectively place pump **58** in fluid communication with a lumen of an aspiration catheter of interventional devices and accessories **48**.

(41) According to some embodiments, operator **11** operates input systems of remote control station **42** to control robotic drive **24** to position an aspiration catheter in a first position with respect to a thrombus within a patient vessel. Specific examples of such positioning involving manipulation of multiple EMDs in a prescribed sequence will be described herein. The aspiration catheter is coupled to vacuum control **59** which in turn is coupled to pump **58** via tubing. Vacuum control **59** prevents fluid communication between the lumen of the aspiration catheter and the lumen of the tubing. Vacuum control **59** may comprise a clamp, a valve or any other one or more suitable devices.

(42) Control computing system **34** controls pump **58** to initiate evacuation of the tubing lumen which is coupled to vacuum control **59**. In some embodiments, control computing system **34** determines that a pressure within the tubing lumen is equal to or less than a target aspiration pressure. In response to the determination, control computing system **34** controls vacuum control

59 to establish fluid communication between the tubing lumen and the aspiration catheter lumen (e.g., by opening the valve, clamp, or other devices of control **59**). This action may serve to pull (i.e., suck) the thrombus into the aspiration catheter.

(43) In further embodiments, control computing system **34** determines (e.g., via communication with pump **58**) that the pressure within the tubing lumen is greater than a second pressure and, in response, controls vacuum control **59** to terminate fluid communication between the tubing lumen and the aspiration catheter lumen. This action may avoid additional aspiration that is not needed to remove the thrombus.

(44) Alternatively, control computing system **34** instructs local control station **38** or remote control station **40** that the pressure within the tubing lumen is equal to or less than the target aspiration pressure, and this information is presented to operator **11** via, e.g., display **30**. Operator **11** may then transmit a command (e.g., using one or more input systems **28**) to control computing system **34** to automatically establish fluid communication between the tubing lumen and the aspiration catheter lumen. Similarly, operator **11** may be presented with an indication that the pressure within the tubing lumen is greater than the second pressure and, in response operator **11** may transmit a command to control computing system **34** to terminate fluid communication between the tubing lumen and the aspiration catheter.

(45) According to still further embodiments, after automatically establishing fluid communication between the tubing lumen and the aspiration catheter, and before determining whether the pressure within the tubing lumen is greater than a second pressure, control computing system **34** may determine that the pressure within the tubing lumen is within a predefined range. If it is determined that the pressure within the tubing lumen is within the predefined range, control computing system **34** may instruct robotic drive **24** to adjust the position of the aspiration catheter lumen (e.g., per a pre-programmed sequence of movements that may or may not take into account intermediate pressure changes) until the pressure within the tubing lumen is less than a threshold aspiration pressure.

(46) Catheter-based procedure system **10** may be connected or configured to include any other systems and/or devices not explicitly shown. For example, catheter-based procedure system **10** may include image processing engines, data storage and archive systems, automatic balloon and/or stent inflation systems, medicine injection systems, medicine tracking and/or logging systems, user logs, encryption systems, or systems to restrict access or use of catheter-based procedure system **10**. It should be noted that any of the determinations attributed herein to control computing system **34** may be performed by any suitable component of or connected to system **10**.

(47) As mentioned, control computing system **34** is in communication with bedside unit **20** which includes robotic drive **24**, positioning system **22** and may include additional controls and displays **46**. Control computing system **34** may receive signals from remote control station **42** based on user manipulation of controls of an input system of remote control station **42**, and may provide corresponding control signals to bedside unit **20** to control the operation of the motors and drive mechanisms used to drive corresponding EMDs in various degrees of freedom. The various drive mechanisms may be provided as part of robotic drive **24**.

(48) FIG. **3** is a perspective view of robotic drive **24** for catheter-based procedure system **10** in accordance with some embodiments. Embodiments are not limited to the robotic drive **24** of FIG. **3**. Robotic drive **24** of FIG. **3** includes multiple device modules **32a-d** coupled to linear member **60**. Each device module **32a-d** is coupled to linear member **60** via respective stage **62a-d** moveably mounted to linear member **60**. A device module **32a-d** may be connected to a stage **62a-d** using a connector such as an offset bracket **78a-d**. In another embodiment, a device module **32a-d** is directly mounted to a stage **62a-d**. Each stage **62a-d** may be independently actuated to move linearly along linear member **60**. Accordingly, each stage **62a-d** (and corresponding device module **32a-d** coupled to stage **62a-d**) may independently move relative to each other and linear member **60**.

(49) A drive mechanism is used to actuate each stage **62a-d**. In the embodiment shown in FIG. 3, the drive mechanism includes independent stage translation motors **64a-d** coupled to each stage **62a-d** and stage drive mechanism **76**, for example, a lead screw via a rotating nut, a rack via a pinion, a belt via a pinion or pulley, a chain via a sprocket, or stage translation motors **64a-d** may be linear motors themselves. In some embodiments, stage drive mechanism **76** may be a combination of these mechanisms, for example, each stage **62a-d** could employ a different type of stage drive mechanism. In some embodiments where the stage drive mechanism is a lead screw and rotating nut, the lead screw may be rotated and each stage **62a-d** may engage and disengage from the lead screw to enable movement thereof, e.g., to advance or retract. In the embodiment shown in FIG. 3, stages **62a-d** and device modules **32a-d** are in a serial drive configuration.

(50) Each device module **32a-d** includes device module **68a-d** and cassette **66a-d** mounted on and coupled to the device module **68a-d**. In the embodiment shown in FIG. 3, each cassette **66a-d** is mounted to device module **68a-d** in a vertical orientation. In other embodiments, each cassette **66a-d** may be mounted to device module **68a-d** in other mounting orientations. Each cassette **66a-d** is configured to interface with and support a proximal portion of an EMD (not shown). In addition, each cassette **66a-d** may include elements to provide one or more degrees of freedom in addition to the linear motion provided by the actuation of the corresponding stage **62a-d** to move linearly along the linear member **60**. For example, a cassette **66a-d** may include elements that may be used to rotate an EMD supported therein when the cassette is coupled to the device module **68a-d**. Each device module **68a-d** includes at least one coupler to provide a drive interface to the mechanisms in each cassette **66a-d** to provide the additional degree of freedom. Each cassette **66a-d** also includes a channel in which a device support **79a-d** is positioned, and each device support **79a-d** is used to prevent an EMD from buckling.

(51) A support arm **77a**, **77b**, and **77c** is attached to each device module **32a**, **32b**, and **32c**, respectively, to provide a fixed point for support of a proximal end of the device supports **79b**, **79c**, and **79d**, respectively. Robotic drive **24** may also include device support connection **72** connected to device support **79**, distal support arm **70** and support arm **770**. Support arm **770** is used to provide a fixed point for support of the proximal end of the distal-most support arm **79a** housed in the distal most device module **32a**. In addition, introducer interface support (redirector) **74** may be connected to device support connection **72** and an EMD (e.g., an introducer sheath). The configuration of robotic drive **24** has the benefit of reducing volume and weight of robotic drive **24** by using actuators on a single linear member.

(52) To prevent contaminating a patient with pathogens, healthcare staff use aseptic technique in a room housing bedside unit **20** and patient **12**. A room housing bedside unit **20** and patient **12** may be, for example, a cath lab or an angio suite. Aseptic technique consists of using sterile barriers, sterile equipment, proper patient preparation, environmental controls and contact guidelines. Accordingly, all EMDs and interventional accessories may be sterilized and allowed contact with either sterile barriers or sterile equipment. In some embodiments, a sterile drape (not shown) is placed over non-sterile robotic drive **24**. Each cassette **66a-d** is sterilized and acts as a sterile interface between draped robotic drive **24** and at least one EMD. Each cassette **66a-d** can be designed to be sterile for single use or to be re-sterilized in whole or part so that a cassette **66a-d** or its components can be used in multiple procedures.

(53) As used herein, the term cassette generally refers to a component of a robotic drive system including components to support and move (e.g., rotate and/or translate) at least one EMD. A device module generally refers to a component of a robotic drive system that includes one or more motors with drive couplers which interface with the EMD-moving elements of the cassette. A cassette may provide a sterile interface between at least one EMD and a device module directly or through a device adapter. The term drive module refers to the combination of a device module and a cassette.

(54) In some embodiments, an EMD is a catheter having a hub at a proximal end of the catheter

and a flexible shaft extending from the hub toward the distal end of the catheter, wherein the shaft is more flexible than the hub. In one embodiment, a catheter includes an intermediary portion that transitions between the hub and the shaft which includes an intermediate flexibility that is less rigid than the hub and more rigid than the shaft. In some embodiments the intermediary portion is a strain relief.

(55) The longitudinal axis of a member (for example, an EMD or other element in the catheter-based procedure system) is the line or axis along the length of the member that passes through the center of the transverse cross section of the member in the direction from a proximal portion of the member to a distal portion of the member. For example, the longitudinal axis of a guidewire is the central axis in the direction from a proximal portion of the guidewire toward a distal portion of the guidewire even though the guidewire may be non-linear in the relevant portion.

(56) Axial movement of a member refers to translation of the member along the longitudinal axis of the member. For example, when the distal end of an EMD is axially moved in a distal direction along its longitudinal axis into or further into the patient, the EMD is being advanced. When the distal end of an EMD is axially moved in a proximal direction along its longitudinal axis out of or further out of the patient, the EMD is being withdrawn.

(57) In this regard, axial insertion refers to inserting a first member into a second member along the longitudinal axis of the second member. For example, an EMD that is axially loaded in a collet is axially inserted in the collet. An example of axial insertion could be referred to as back loading a catheter on the proximal end of a guidewire. Lateral insertion refers to inserting a first member into a second member along a direction in a plane perpendicular to the longitudinal axis of the second member. Lateral insertion can also be referred to as radial loading or side loading.

(58) Rotational movement of a member refers to the change in angular orientation of the member about the local longitudinal axis of the member. For example, rotational movement of an EMD corresponds to clockwise or counterclockwise rotation of the EMD about its longitudinal axis due to an applied torque. Continuous motion refers to motion that does not require a reset and is uninterrupted, while discrete motion refers to motion that requires a reset and is interrupted.

(59) The terms distal and proximal define relative locations of two different features. With respect to a robotic drive, the terms distal and proximal are defined by the position of the robotic drive in its intended use relative to a patient.

(60) When used to define a relative position, the distal feature is the feature of the robotic drive that is closer to the patient than a proximal feature when the robotic drive is in its intended in-use position. Within a patient, any vasculature landmark further away along the path from the access point is considered more distal than a landmark closer to the access point, where the access point is the point at which the EMD enters the patient. Similarly, the proximal feature is the feature that is farther from the patient than the distal feature when the robotic drive is in its intended in-use position.

(61) When used to define direction, the distal direction refers to a path on which something is moving or is aimed to move or along which something is pointing or facing from a proximal feature toward a distal feature and/or patient when the robotic drive is in its intended in-use position. The proximal direction is the opposite direction of the distal direction. For example, referring to FIG. 1, a robotic device is shown from the viewpoint of an operator facing a patient. In this arrangement, the distal direction is along the positive X coordinate axis and the proximal direction is along the negative X coordinate axis.

(62) With respect to movement of modules, and referring to FIG. 3, an EMD is moved in a distal direction on a path toward a patient through introducer interface support 74 which defines the distal end of robotic drive 24. The proximal end of robotic drive 24 is the point furthest from the distal end along the negative X axis.

(63) With respect to positions of the individual modules, and also referring to FIG. 3, the most distal device module is device module 32a closest to the distal end of robotic drive 24. The most proximal device module is device module 32d positioned furthest from the distal end of robotic

drive **24** along the negative X axis. The relative position of device modules is determined by their relative location to the distal end of the robotic drive. For example, device module **32b** is distal to device module **32c**.

(64) With respect to distal/proximal portions, sections or ends of an EMD or the robotic drive, the portions of cassette **66a** and device module **68a** are defined by their relative location to the distal end of the robotic drive. For example, the distal end of cassette **66a** is the portion of the cassette that is closest to the distal end of the robotic drive and the proximal end of cassette **66a** is the portion of the cassette that is furthest from the distal end of the robotic drive along the negative X axis when the cassette is in-use position on device module **68a**. Stated in another way, the distal end of cassette **66a** is the portion of the cassette through which an EMD is closest to the path leading to a patient in the in-use position.

(65) As previously discussed, embodiments of a control station such as control station **26** may include a variety of different input systems for controlling bedside unit **20**. Input systems can include a variety of different input controls (e.g., buttons, scroll wheels, joysticks, touch screens) that can be manipulated by a user to control (or, instruct) operation of robotic drive **24**. These input controls can be arranged in different layouts or patterns on the input system to facilitate desired functions and cooperative sequencing thereof to perform a desired task requiring independent (and sometimes simultaneous) movement of multiple EMDs.

(66) Additionally, embodiments of an input system can be configured to operate in a variety of different control modes. The functions assigned to one or more controls of an input system in a first control mode may differ from the functions assigned to the one or more controls in a second control mode, and control modes may be selected based on the procedure being performed, the device or devices to be controlled, operator preferences, or any other factors. An input system can be configured to switch between different control modes in response to input from an operator or from control computing system **34**.

(67) An input system as described herein may be fixed to, integrated with, or simply rested atop a surface of control station **26**. As described herein, an input system may comprise a single integrated housing or multiple independently-movable housings.

(68) FIG. **4** is a schematic diagram of a thrombectomy system according to some embodiments. Each component of system **400** may be implemented by one or more devices comprised of any suitable combination of hardware and/or software, and two or more illustrated components may be implemented by a same device(s).

(69) FIG. **4** shows vessel **410**, which may comprise a vein or artery internal to a patient body. Vessel **420** may be located at any position within a patient. Thrombus **420** is disposed within vessel **410**. Embodiments are not limited to a thrombus and vessel of relative sizes and shapes shown in FIG. **4** or in any subsequently-described figures. Embodiments are also not limited to a single substantially-contiguous thrombus.

(70) Catheter control **430** operates to navigate catheter **440** to thrombus **420** within vessel **410**. As shown, catheter control **430** has maneuvered tip **440a** of catheter **440** to a position adjacent to thrombus **420** within vessel **410**. Catheter **440** may comprise any EMD defining a lumen in which negative pressure may be generated. Catheter **440** may comprise a purpose-designed aspiration catheter defining a catheter lumen.

(71) As will be described below, catheter control **430** may manipulate (i.e., advance/retract/rotate) one or more unshown guidewires and catheters in order to navigate catheter **440** to the illustrated position. Such navigation may be facilitated using a robotic drive such as but not limited to robotic drive **24**. Accordingly, in some embodiments, catheter control **430** comprises catheter-based procedure system **10**.

(72) Vacuum source **450** may comprise a pump or other device suitable for generating negative pressure in an interventional theatre. According to system **400**, vacuum source **450** is coupled to tubing **460a** which is coupled to valve/clamp **470**. Valve/clamp **470** is coupled to tubing **460b**

which is coupled to catheter **440** via catheter control **430**. Tubing **460a** and tubing **460b** may comprise any tubing suitable for thrombus aspiration as described therein, and together comprise a tubing lumen.

(73) When valve/clamp **470** is closed, negative pressure generated by vacuum source **450** exists within a lumen of tubing **460a** but not within a lumen of tubing **460b** (and therefore not within catheter **440**). When valve/clamp **470** is open, vacuum source **450**, the lumen of tubing **460a**, the lumen of tubing **460b** and the lumen of catheter **440** are all in fluid communication with one another. Accordingly, any negative pressure generated by vacuum source **450** is experienced at tip **440a** of catheter **440**.

(74) According to system **400**, pressure sensor **480** is coupled to or integrated with vacuum source **450**. Pressure sensor **480** may determine a pressure generated by vacuum source **450**, for example a pressure generated within tubing **460a** while valve/clamp **470** is closed. Valve/clamp **470** may monitor this pressure via communication with pressure sensor **480** and determine whether to open or close itself based thereon.

(75) For example, in some embodiments, catheter control **430** manipulates a series of EMDs under control of an operator to position tip **440a** of catheter **440** adjacent to thrombus **420**. Next, tubing **460b** is coupled, either manually or via mechanisms of catheter control **430**, to catheter **440** such that tubing **460b** and catheter **440** are in fluid communication with one another. Vacuum source **450** is operated (e.g., via a command from catheter control **430** or an operator) to begin generation of negative pressure within tubing **460a**.

(76) Valve/clamp **470** monitors pressure determined by pressure sensor **480** until it is determined that a pressure within tubing **460a** is equal to or less than a target aspiration pressure. Alternatively, pressure sensor **480** may provide a signal to valve/clamp **470** once pressure sensor **480** determines that a pressure within tubing **460a** is equal to or less than a target aspiration pressure. In either case, in response to the determination, valve/clamp **470** is opened to automatically establish fluid communication between tubing **460a**, tubing **460b** and catheter **440**. Consequently, thrombus **420** may be forcefully removed from vessel **410** and drawn into catheter **440**.

(77) After opening of valve/clamp **470**, valve/clamp **470** may further determine, based on pressure sensor **480**, that the pressure within tubing lumen **460a** is greater than a second pressure and, in response, close itself to automatically terminate fluid communication between tubing lumen **460a** and catheter **440**. Such a procedure may serve to detect completion of the removal of thrombus **420** and to accordingly terminate aspiration.

(78) In still further embodiments, after opening of valve/clamp **470** but before closing valve/clamp **470** because the pressure within tubing lumen **460a** is greater than a second pressure, valve/clamp **470** determines based on pressure sensor **480** that the pressure within tubing lumen **460a** is within a predefined range. The predefined range is higher than the first pressure and lower than the second pressure and is intended to indicate the existence of some resistance to suction into catheter **440** but not to the extent expected if tip **440a** were positioned against thrombus **420**. If it is determined that the pressure within tubing lumen **460a** is within the predefined range, valve/clamp **470** (or pressure sensor **480**, for example) may communicate with catheter control **430** to change the position (e.g., rotate) catheter **440** until the pressure within tubing lumen **460a** is equal to or less than a desired pressure.

(79) In some embodiments, valve/clamp **470** is a component of catheter control **430**. Accordingly, a single section of tubing might extend from vacuum source **450** to such a valve/clamp **470**, which is then directly coupled to catheter **440** by catheter control **430**.

(80) FIG. 5 is a schematic diagram of a thrombectomy system according to some embodiments. System **500** includes vessel **510** and thrombus **520** disposed within vessel **510** as described above. Catheter control **530** operates to navigate catheter **540** to thrombus **520** as described with respect to catheter control **430** and catheter **440**.

(81) Vacuum source **550**, tubing **560a**, tubing **560b**, valve/clamp **570** and pressure sensor **580** may

be configured as described above with respect to similarly-named elements of system **400**. System **500** also includes pressure control **590** disposed between pressure sensor **580** and valve/clamp **570**. Pressure control **590** may execute the determinations and control of valve/clamp **570** which was attributed above to valve/clamp **470**. Generally, pressure control **590** may monitor the pressure within tubing lumen **560a** via communication with pressure sensor **580** and instruct valve/clamp **570** to open or close based thereon. In some embodiments, pressure control **590** is a component of catheter control **530**. In this regard, pressure control **590** may comprise control computing system **34** of system **10**.

(82) Pressure control **590** may initiate operation of vacuum source **550** to begin generation of negative pressure within tubing **560a**. Pressure control **590** monitors pressure determined by pressure sensor **580** until a pressure within tubing **560a** is equal to or less than a target aspiration pressure. In response to this determination, pressure control **590** instructs valve/clamp **570** to open to automatically establish fluid communication between tubing **560a**, tubing **560b** and catheter **540**.

(83) Subsequent to this opening of valve/clamp **570**, pressure control **590** may further monitor pressure determined by pressure sensor **580** until a pressure within tubing **560a** is greater than a second pressure and, in response, instruct valve/clamp **570** to close itself to automatically terminate fluid communication between tubing lumen **560a** and catheter **540**. Additionally, if pressure control **590** determines that the pressure within tubing lumen **560a** is within a predefined range higher than the first pressure and lower than the second pressure, pressure control **590** may communicate with catheter control **530** to change the position of (e.g., rotate) catheter **540** until the pressure within tubing lumen **560a** is equal to or less than the first pressure.

(84) FIG. **6** is a schematic diagram of a thrombectomy system according to some embodiments. System **600** is similar to system **500** of FIG. **5**, except that the above-described functions of pressure sensor **580** and pressure control **590** are combined within pressure sensor and control **690**. Vacuum source **680**, tubing **660a**, tubing **660b**, valve/clamp **670** and catheter control **630** may be configured as described above with respect to similarly-named elements of system **500**.

(85) Pressure sensor and control **690** may monitor the pressure within tubing lumen **660a** via communication with vacuum source **680** and instruct valve/clamp **670** to open or close based thereon. As mentioned with respect to pressure control **590** of system **500**, in some embodiments, pressure sensor and control **690** is a component of catheter control **630**. For example, pressure sensor and control **690** may comprise control computing system **34** of system **10**.

(86) Pressure sensor and control **690** may initiate operation of vacuum source **680** to begin generation of negative pressure within tubing **660a**. Pressure sensor and control **690** monitors pressure generated by vacuum source **680** (e.g., by communication with vacuum source **680** which includes a pressure sensor, by optical sensing of tubing **660a**) until it is determined that a pressure within tubing **660a** is equal to or less than a target aspiration pressure. In response to this determination, pressure sensor and control **690** instructs valve/clamp **670** to open to automatically establish fluid communication between tubing **660a**, tubing **560b** and catheter **640**.

(87) Next, pressure sensor and control **690** may determine that a pressure within tubing **660a** has become greater than a second pressure and, in response, instruct valve/clamp **670** to close itself to automatically terminate fluid communication between tubing lumen **660a** and catheter **640**. If, between the opening and closing of valve/clamp **670** as described above, pressure sensor and control **690** determines that the pressure within tubing lumen **660a** is within a predefined range higher than the first pressure and lower than the second pressure, pressure sensor and control **690** may communicate with catheter control **630** to change the position of (e.g., rotate) catheter **640** until the pressure within tubing lumen **660a** is equal to or less than the first pressure.

(88) FIG. **7** is a schematic diagram of a thrombectomy system according to some embodiments. System **700** includes presentation of pressure-based indications to an operator, and subsequent operator control of system **700**. System **700** is presented in two alternative arrangements, a first in which pressure sensor **770a** is coupled to or integrated with vacuum source **740**, and a second in

which pressure sensor **770b** is separate from vacuum source **740**. Pressure sensor **770b** may determine a pressure within tubing **750a** via optical means, a sensor element within tubing **750a**, or any other suitable technique.

(89) Regardless of the implementation of the pressure sensor (e.g., pressure sensor **770a** or pressure sensor **770b**), system **700** includes presentation of an indication of the pressure within tubing **750a** to operator **775** via display **780**. Although FIG. 7 shows a direct communication link between pressure sensor **770a**/pressure sensor **770b** and display **780**, the pressure may be transmitted from pressure sensor **770a**/pressure sensor **770b** to display **780** via any number of intermediary components, such as, for example, control computing system **34**.

(90) In operation, operator **775** operates catheter control **730** (e.g., of a catheter-based robotic procedure system such as system **10**) to position catheter **740** with respect to thrombus **720** within vessel **710**. Once the position is satisfactory to operator **775**, operator **775** initiates operation of vacuum source **740** to begin generation of negative pressure within tubing **750a**.

(91) Display **780** presents an indication of the negative pressure to operator **775**. The indication may comprise a pressure value and/or an indication that a pressure within tubing **750a** is equal to or less than a target aspiration pressure. The indication may simply comprise a notification, presented once a pressure within tubing **750a** is equal to or less than the target aspiration pressure, that the system is ready for aspiration. After receiving the indication, operator **775** may manipulate operator control **790** to instruct valve/clamp **760** to open and thereby automatically establish fluid communication between tubing **750a**, tubing **750b** and catheter **740**. Operator control **790** may comprise any of the input systems described herein, including but not limited to a touch screen (e.g., display **780** may also comprise operator control **790**), a console button, a foot switch, and a joystick.

(92) After opening of valve/clamp **760**, display **780** may further present an indication that a pressure within tubing **750a** has become greater than a second pressure. The indication may comprise a notification to terminate aspiration. In response to the indication, operator **775** may manipulate operator control **790** to close valve/clamp **760** to terminate fluid communication between tubing **750a** and catheter **740**. In some embodiments of system **700**, the pressure within tubing **750a** is monitored and, once it is determined that the pressure within tubing **750a** has become greater than a second pressure, valve/clamp **760** is automatically instructed to close without intervention of operator **775**.

(93) Between the aforementioned opening and closing of valve/clamp **760**, display **780** may present an indication that the pressure within tubing lumen **750a** is within a predefined range higher than the first pressure and lower than the second pressure. The indication may comprise a pressure value and/or a notification to reposition catheter **740** with respect to thrombus **720**. Based on the indication, operator **775** may instruct operation of catheter control **730** to reposition catheter **740**. The repositioning may comprise a pre-programmed set of movements and/or individual movements instructed by operator **775**. In some embodiments, display **780** presents an indication once the pressure within tubing lumen **750a** has decreased to equal to or less than the first pressure, at which point operator **775** may cease instruction of catheter control **730**.

(94) FIG. 8 is a schematic diagram of a thrombectomy system according to some embodiments. System **800** may comprise an implementation of system **700**. In particular, valve **840** comprises an implementation of valve/clamp **760**. Similar to the description of system **700**, operator **875** may operate catheter control **820** to position catheter **810** with respect to a thrombus within a vessel. Once the position is satisfactory, vacuum source **830** is operated to begin generation of negative pressure within tubing **850a**.

(95) Display **870** presents an indication of the negative pressure to operator **875**. After receiving the indication, operator **875** may directly manipulate valve **740** to open and thereby automatically establish fluid communication between tubing **850a**, tubing **850b** and catheter **810**. In the illustrated embodiment, valve **840** comprises a flow control valve including slider switch **845**

which may be operated using one hand to selectively open or close valve **840**. Embodiments may employ any suitable device for establishing and terminating fluid communication between two lumens.

(96) Display **780** may further present an indication that a pressure within tubing **750a** has become greater than a second pressure. The indication may comprise a notification to terminate aspiration. In response to the indication, operator **775** may manipulate operator control **790** to close valve/clamp **760** to terminate fluid communication between tubing **750a** and catheter **740**. In some embodiments of system **700**, the pressure within tubing **750a** is monitored and, once it is determined that the pressure within tubing **750a** has become greater than a second pressure, valve/clamp **760** is automatically instructed to close without intervention of operator **775**.

(97) Between the aforementioned opening and closing of valve/clamp **760**, display **780** may present an indication that the pressure within tubing lumen **750a** is within a predefined range higher than the first pressure and lower than the second pressure. The indication may comprise a pressure value and/or a notification to reposition catheter **740** with respect to thrombus **720**. Based on the indication, operator **775** may instruct operation of catheter control **730** to reposition catheter **740**. The repositioning may comprise a pre-programmed set of movements and/or individual movements instructed by operator **775**. In some embodiments, display **780** presents an indication once the pressure within tubing lumen **750a** has decreased to equal to or less than the first pressure, at which point operator **775** may cease instruction of catheter control **730**.

(98) FIG. **9** comprises a flow diagram of process **900** to aspirate a thrombus according to some embodiments. Process **900** and other processes described herein may be executed using any suitable combination of hardware and software. Software program code embodying these processes may be stored by any non-transitory tangible medium, including a fixed disk, a volatile or non-volatile random-access memory, a DVD, a Flash drive, and a magnetic tape, and executed by any suitable processing unit, including but not limited to one or more microprocessors, microcontrollers, processing cores, and processor threads. Embodiments are not limited to the examples described below.

(99) Initially, at **S910**, an aspiration catheter defining an aspiration catheter lumen is positioned in a first position with respect to a thrombus. For example, an operator may control a robotic drive to manipulate a series of EMDs in a prescribed sequence at **S910** to position a tip of an aspiration catheter against a thrombus. According to one such prescribed sequence, an introducer sheath is inserted into an access site, such as but not limited to a radial or femoral artery. Next, a diagnostic guidewire is inserted into the sheath and is advanced to the base of the skull by a robotic drive in response to operator commands. A guide catheter and base catheter are then advanced (either simultaneously or sequentially) over the diagnostic guidewire to the base of the skull. The diagnostic guidewire and guide catheter are then removed, leaving the base catheter.

(100) A coaxial “stack” consisting of a microwire surrounded by a microcatheter surrounded by an aspiration catheter is then inserted into the base catheter. With the microwire leading, the stack passes the base of the skull (i.e., where the base catheter terminates) and navigates the vasculature until the microwire reaches the thrombus. The aspiration catheter is then advanced, either with or without the microcatheter (i.e., for structural support), until a lumen of the aspiration catheter is in contact with the thrombus. The foregoing process is aided by contemporaneously-acquired and contrast enhanced X-ray images.

(101) FIG. **10** illustrates vessel **1000** in which base catheter **1010** resides. A stack consisting of microwire **1020**, microcatheter **1030** and aspiration catheter **1040** has passed through base catheter **1010** on the way to thrombus **1050**. Microwire **1020** has reached thrombus **1050** and microcatheter **1030** is being advanced over microwire **1020** toward thrombus **1050**. Next, aspiration catheter **1040** will be advanced over microcatheter **1030** (and microwire **1020**) until aspiration catheter **1040** reaches thrombus **1050**. Microwire **1020** and microcatheter **1030** may then be removed from within aspiration catheter **1040**.

(102) Returning to process **900**, evacuation of a tubing lumen to generate negative pressure therein is initiated at **S920**. The tubing lumen may be coupled to a pump or other vacuum source as noted above. As also noted, the tubing lumen is not in fluid communication with the aspiration catheter. According to some embodiments, an operator operates an input control associated with a robotic catheter-based procedure system to instruct operation of the vacuum source at **S920**.

(103) Flow pauses at **S930** until the pressure within the tubing lumen is equal to (or less than) a target pressure. The target pressure may be pre-defined and/or set by the operator, and may be determined based on the size and/or shape of the thrombus, the nature of the vessel (e.g., delicate, tortuous, damaged) in which the thrombus resides, or any other factor. Flow proceeds to **S940** once it is determined that a pressure within the tubing lumen equal to (or less than) the target pressure.

(104) At **S940**, fluid communication is automatically established between the tubing lumen and the aspiration catheter lumen. Several examples of **S940** are described herein. In one example, both the tubing and the aspiration catheter are coupled to either end of a closed valve prior to **S940**, and **S940** includes opening the valve. The valve may be opened via an electromechanical actuator or manually by an operator. Implementations of **S940** are not limited to those described herein.

(105) **S1110** through **S1140** of process **1100** of FIG. **11** may proceed similarly to **S910** through **S940** of process **900**. Flow pauses after **S1140**, at **S1150**, until it is determined that the pressure within the tubing lumen is greater than (or equal to) a second target pressure. As described above, the second target pressure may be a pressure that is expected in a case that the thrombus has been evacuated from the vessel (and the aspiration catheter).

(106) Flow proceeds to **S1160** once the second target pressure is reached. At **S1160**, the fluid communication between the tubing lumen and the aspiration catheter lumen is terminated. Continuing the above example, **S1160** may comprise electromechanical or manual closing of a valve disposed between the tubing lumen and the aspiration catheter lumen.

(107) Turning to FIGS. **12**, **S1205**, **S1210** and **S1215** of process **1200** may proceed similarly to respective ones of **S910**, **S920** and **S930** of process **900**. At **S1220**, an indication is presented to an operator, indicating that the pressure within the tubing lumen (i.e., the pressure to be experienced by the aspiration catheter lumen) has reached the target pressure. As described above, the indication may be presented via a display of an operator console, and may be communicated from the vacuum source or a separate pressure sensor to a control computing system which then instructs presentation of the indication.

(108) Flow pauses at **S1225** until a command from the operator is received. The target pressure may be maintained during the pause at **S1225**. The operator may operate an input control of a control console to instruct the system to perform aspiration, in response to which flow proceeds to **S1230**.

(109) Fluid communication is automatically established between the tubing lumen and the aspiration catheter lumen at **S1230**. Flow then pauses at **S1235** until it is determined that the pressure within the tubing lumen is greater than (or equal to) a second target pressure. Another indication is presented to the operator at **S1240**, indicating that the pressure within the tubing lumen (and within the aspiration catheter lumen) has reached a second target pressure. Flow pauses at **S1245** until a next command from the operator is received.

(110) The operator may operate an input control of the control console at **S1245** to instruct the system to terminate aspiration. In response, at **S1250**, the fluid communication between the tubing lumen and the aspiration catheter lumen is terminated. For example, a valve disposed between the tubing lumen and the aspiration catheter lumen may be closed at **S1250**.

(111) FIG. **13** is a flow diagram of process **1300** according to some embodiments. **S1310** through **S1340** of process **1300** may proceed similarly to **S910** through **S940** of process **900**. Flow proceeds to **S1350** after fluid communication between a tubing lumen and an aspiration catheter is established at **S1340**.

(112) At **S1350**, it is determined whether the pressure within the tubing lumen is within a

predefined range. As described above, the predetermined range may comprise a predetermined range of pressures which indicate that the aspiration catheter is not engaged against the thrombus in a manner providing suitable suction thereto. Flow proceeds to **S1370** if the pressure within the tubing lumen is not within the predefined range.

(113) If it is determined at **S1350** that the pressure within the tubing lumen is within the predefined range, the position of the aspiration catheter lumen is automatically adjusted at **S1360** until the pressure within the tubing lumen is less than a threshold aspiration pressure. This adjustment may be automatically performed without operator intervention using a set of one or more catheter movements intended to better-position the catheter. In some embodiments, the adjustment is performed under partial or full operator control.

(114) At **S1370**, it is determined whether the pressure within the tubing lumen is greater than a second pressure. If not, flow returns to **S1350** and continues as described above. If so, fluid communication between the tubing lumen and the aspiration catheter lumen is automatically terminated at **S1380**, using any system described herein or that is known.

(115) Those in the art will appreciate that various adaptations and modifications of the above-described embodiments can be configured without departing from the claims. Therefore, it is to be understood that the claims may be practiced other than as specifically described herein.

Claims

1. A method comprising: receiving a first command to position a catheter lumen in a position with respect to a thrombus, the catheter lumen defined by a catheter; and while the catheter lumen is positioned in the position with respect to the thrombus: receiving a second command to initiate evacuation of a tubing lumen while the tubing lumen is not in fluid communication with the catheter lumen; determining while the tubing lumen is not in fluid communication with the catheter lumen, that a pressure within the tubing lumen is equal to or less than a target aspiration pressure; and in response to the determination that the pressure within the tubing lumen is equal to or less than the target aspiration pressure while the tubing lumen is not in fluid communication with the catheter lumen, automatically establishing fluid communication between the tubing lumen and the catheter lumen.
2. A method according to claim 1, further comprising: after automatically establishing fluid communication between the tubing lumen and the catheter, determining whether the pressure within the tubing lumen is greater than a second pressure; and if it is determined that the pressure within the tubing lumen is greater than the second pressure, automatically terminating fluid communication between the tubing lumen and the catheter lumen.
3. A method according to claim 2, further comprising: after automatically establishing fluid communication between the tubing lumen and the catheter and before determining whether the pressure within the tubing lumen is greater than the second pressure, determining whether the pressure within the tubing lumen is within a predefined range; and if it is determined that the pressure within the tubing lumen is within the predefined range, automatically adjusting the position of the catheter lumen until the pressure within the tubing lumen is less than a threshold aspiration pressure.
4. A method according to claim 1, further comprising: after automatically establishing fluid communication between the tubing lumen and the catheter lumen, determining whether the pressure within the tubing lumen is within a predefined range; and if it is determined that the pressure within the tubing lumen is within the predefined range, automatically adjusting the position of the catheter lumen until the pressure within the tubing lumen is less than a threshold aspiration pressure.
5. A method according to claim 1, further comprising: after automatically establishing fluid communication between the tubing lumen and the catheter lumen, determining whether the

pressure within the tubing lumen is greater than a second pressure; and if it is determined that the pressure within the tubing lumen is greater than the second pressure, presenting an indication to an operator; after presenting the indication, and while the pressure within the tubing lumen is greater than the second pressure, receiving a command from the operator; and in response to receipt of the command, automatically terminating fluid communication between the tubing lumen and the catheter lumen.

6. A method according to claim 5, further comprising: after automatically establishing fluid communication between the tubing lumen and the catheter lumen and before determining that the pressure within the tubing lumen is greater than the second pressure, determining whether the pressure within the tubing lumen is within a predefined range; and if it is determined that the pressure within the tubing lumen is within the predefined range, automatically adjusting the position of the catheter lumen until the pressure within the tubing lumen is less than a threshold aspiration pressure.

7. A method according to claim 1, wherein automatically establishing fluid communication between the tubing lumen and the catheter lumen comprises: presenting a first indication to an operator; after presenting the first indication, and while the pressure within the tubing lumen is equal to or less than the target aspiration pressure, receiving a third command from the operator to establish fluid communication between the tubing lumen and the catheter lumen; and in response to receipt of the third command, automatically establishing fluid communication between the tubing lumen and the catheter lumen.

8. A method according to claim 7, further comprising: after automatically establishing fluid communication between the tubing lumen and the catheter lumen, determining whether the pressure within the tubing lumen is greater than a second pressure; and if it is determined that the pressure within the tubing lumen is greater than the second pressure, presenting a second indication to the operator; after presenting the second indication, and while the pressure within the tubing lumen is greater than the second pressure, receiving a fourth command from the operator; and in response to receipt of the fourth command, automatically terminating fluid communication between the tubing lumen and the catheter lumen.

9. A method according to claim 8, further comprising: after automatically establishing fluid communication between the tubing lumen and the catheter lumen and before determining that the pressure within the tubing lumen is greater than the second pressure, determining whether the pressure within the tubing lumen is within a predefined range; and if it is determined that the pressure within the tubing lumen is within the predefined range, automatically adjusting the position of the catheter lumen until the pressure within the tubing lumen is less than a threshold aspiration pressure.

10. A method according to claim 8, further comprising: after automatically establishing fluid communication between the tubing lumen and the catheter lumen and before determining that the pressure within the tubing lumen is greater than the second pressure, determining whether the pressure within the tubing lumen is within a predefined range; and if it is determined that the pressure within the tubing lumen is within the predefined range, presenting a third indication to the operator to adjust the position of the catheter lumen until the pressure within the tubing lumen is less than a threshold aspiration pressure.

11. A method according to claim 7, further comprising: after automatically establishing fluid communication between the tubing lumen and the catheter lumen, determining whether the pressure within the tubing lumen is within a predefined range; and if it is determined that the pressure within the tubing lumen is within the predefined range, automatically adjusting the position of the catheter lumen until the pressure within the tubing lumen is less than a threshold aspiration pressure.

12. A method according to claim 7, further comprising: after automatically establishing fluid communication between the tubing lumen and the catheter lumen, determining whether the

pressure within the tubing lumen is within a predefined range; and if it is determined that the pressure within the tubing lumen is within the predefined range, presenting a second indication to the operator to adjust the position of the catheter lumen until the pressure within the tubing lumen is less than a threshold aspiration pressure.
